WIC Self-Learning Report

Learning the Basics of Fusion 360 for a Solenoid-Driven Robotic Piano Player

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Project: Robotic piano player using solenoids

Assignment alignment: Mechanical and Manufacturing Engineering Topic (3) - Learn the basics of Fusion 360

Date: April 13, 2026

This report documents a self-directed learning activity that most closely matches the course example "Learn the basics of Fusion 360." The training was completed through YouTube and Autodesk Fusion tutorials, then applied to the capstone design process.

Self-Learning Report

Capstone project	Autonomous robotic piano player that uses solenoids to actuate piano keys
Skill learned	Basics of Autodesk Fusion 360 for 3D modeling and design iteration
Training time	Approximately 7 hours of self-directed instruction before additional practice and project use
Primary sources	Four Product Design Online YouTube lessons plus Autodesk Fusion tutorials and help resources
Project impact	Supported frame design, design iteration, and stress/loading considerations related to the solenoids and frame

1. Skill Learned and Why I Chose It

The self-directed learning topic that I have selected is the basics of Autodesk Fusion 360. Although there are multiple examples in the class syllabus, this example aligns most with "Learn the basics of Fusion 360" since the learning occurred via YouTube tutorials and Autodesk help files rather than LinkedIn Learning. Prior to undertaking this task, my experience in 3D modeling was limited, which meant that Fusion 360 remained unfamiliar to me in practical terms as an engineer. At best, I knew what computer-aided design was capable of doing, though I did not yet have sufficient proficiency in it to make parts quickly and efficiently.

Fusion 360 was selected because I am designing a capstone robot that plays a piano using solenoids; it involves making numerous design choices regarding geometry and frame mounting. It became necessary to model various designs quickly, evaluate different design choices visually, and communicate them with others. As such, learning this software would enable me to acquire some attributes in design that will come in handy during the capstone process, such as spatial reasoning, digital prototyping, design communication, iterative design thinking, and relating geometry to the loading and manufacturing considerations.

2. Self-Learning Training and Evidence

Over the period of seven hours, I taught myself how to use Autodesk Fusion 360 by going through video tutorials and training materials published by Autodesk. The primary path in training I took was the beginner level Product Design Online sequence of tutorials. Instead of following the video lessons passively, I worked inside Fusion 360 on each step shown. This helped me familiarize myself with interface navigation, sketching techniques, feature tree, and difference between designing the model and conceptualizing the object. I additionally went through Autodesk Fusion 360 tutorials, which helped me get accustomed with the fundamentals of the software.

Each tutorial had more advanced models for practice than the previous one. Firstly, I learned sketching, extrude, shell, and pattern features by modeling a Lego handle-like object. Then, I learned how to include a picture into the design, use splines, and create a revolve body by modeling a glass bottle. Then, by modeling a paper clip, I familiarized myself with sweep and tangent arc tool functions. Lastly, I modeled another bottle with the help of lofts and construction planes. These

practice models are depicted in Fig. 1. These documents are proof that I did not merely watch tutorial videos but designed these models.

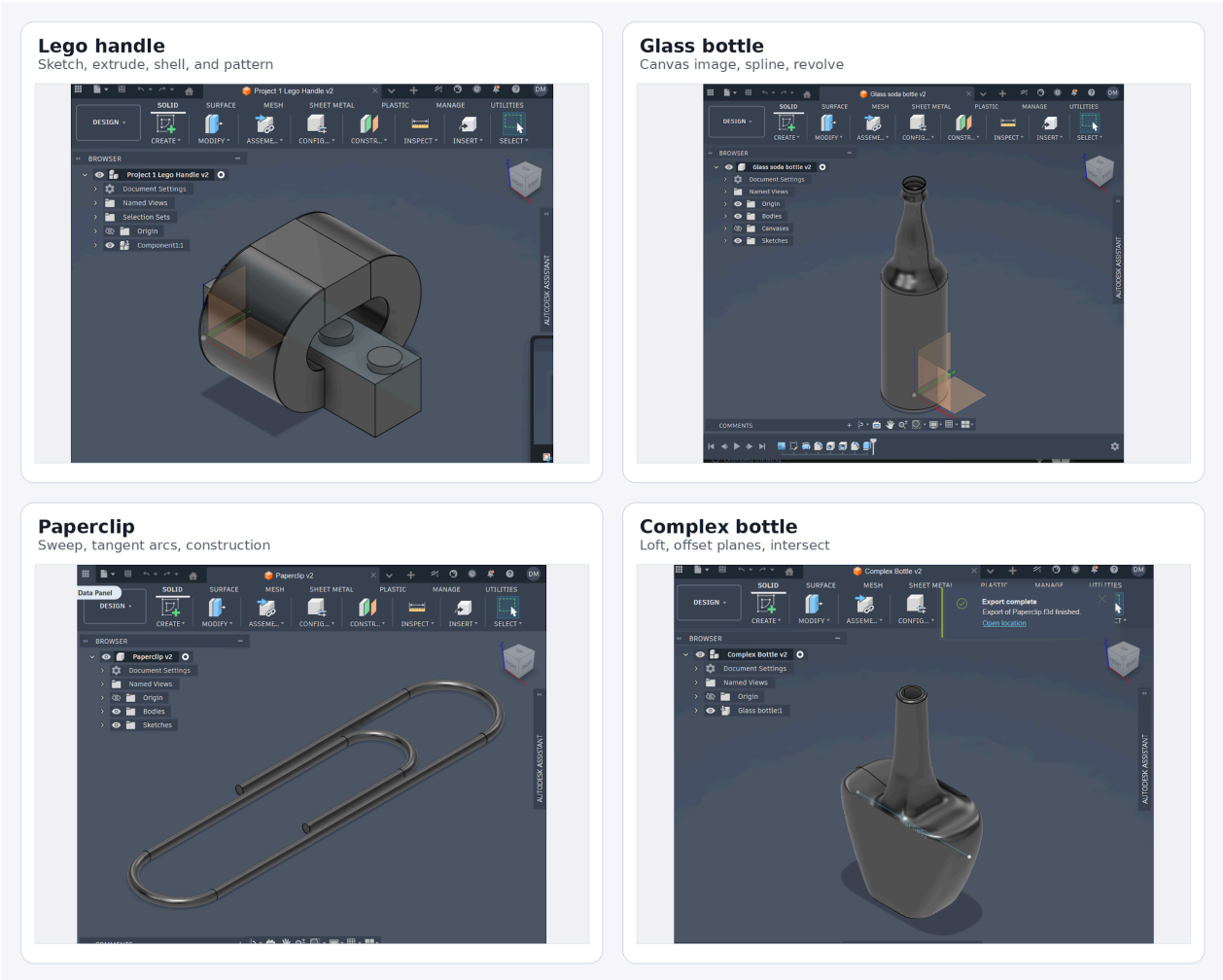

Figure 1. Practice models completed during self-learning in Fusion 360: Lego handle, glass bottle, paperclip, and complex bottle.

3. Application to the Capstone Project

Following the guided practice, I was able to use the same process on the robotic piano player. In this particular project, Fusion 360 was helpful in allowing me to have an easy way to construct and modify the frame digitally as compared to relying solely on trial and error to achieve success. It helped us to model the geometry of the frame digitally, helped us keep track of the design as it evolved, and assisted in thinking through the loading process for the solenoids. In this project, that is especially important since any change in geometry could impact alignment, package size, stiffness, and ease of assembly. Figure 2 shows the Fusion 360 assembly model and drawing package that I developed for the robotic piano player. These project files allowed me to evaluate the frame layout, the placement of the solenoid bank, and the overall assembly dimensions before fabrication.

The greatest change in ability came in terms of efficiency. Once I had mastered the basics of the program, I could go from conception to model in seconds, something I never could do before. This meant that instead of guessing and testing the shape and size of each component until finding an

acceptable configuration, I could design it virtually first and modify it accordingly. Not only was this process more efficient as far as building and testing was concerned, but it was beneficial in improving communication in the project overall, as models are easier to talk about than ideas.

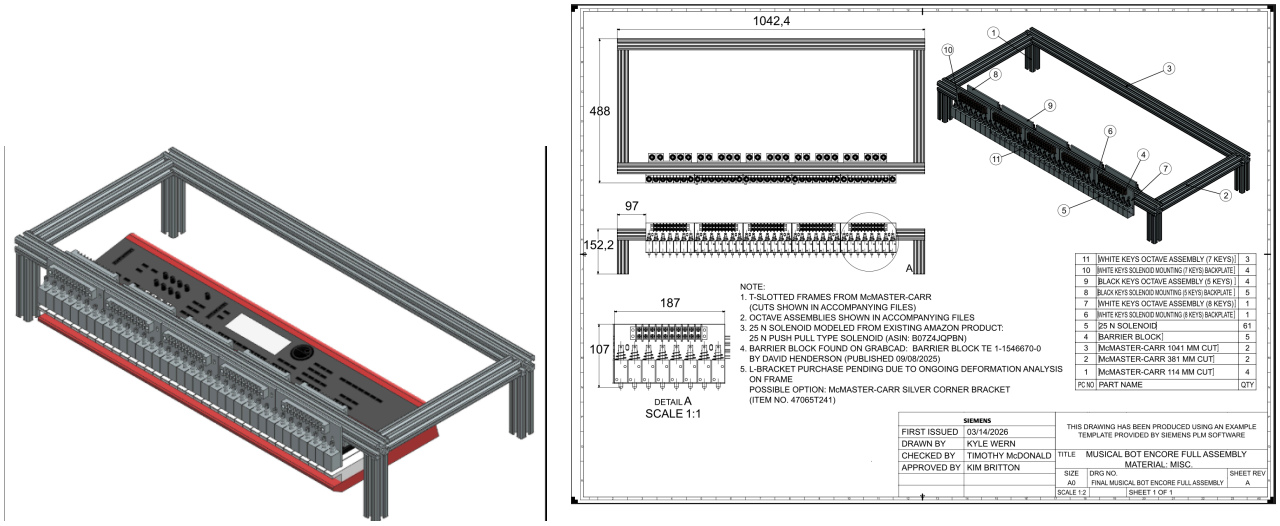

Figure 2. Fusion 360 application to the capstone project: (left) CAD assembly of the robotic piano player showing the frame, keyboard support structure, and solenoid mounting layout; (right) engineering drawing generated from the model showing dimensions, part relationships, and assembly details used for design review and refinement.

4. Results and Design Attributes Developed

The primary outcome of the learning process has been making Fusion 360 a viable software tool to employ on my capstone project as opposed to barely knowing what it looked like or being capable of launching it properly. As shown in Fig. 2, the new Fusion 360 skills were applied to both 3D assembly modeling and technical drawing creation, which made the design easier to evaluate, communicate, and refine before building. This allowed for designing faster while eliminating any unnecessary assumptions and providing better grounds to iterate prior to fabricating or testing anything. Moreover, this process helped me adopt certain mindsets regarding the design process, such as focusing on proper part assembly, load transfer through a structure, and effective changes in case the concept should evolve. All of those skills could immediately apply in practice.

Overall, I have managed to improve my competencies in 3D modeling, making decisions based on my designs, iteration, and linking my choices with performance considerations. These skills are particularly important for a solenoid-powered piano player since there would be a lot of iterations involved in order to develop a viable concept. Mastering Fusion 360 will undoubtedly help me perform better at my future capstone and in any position related to modeling and design.

References

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